

Education

- **Ph.D. in Computer Science**, George Mason University (Aug 2021 – Present)
 - Expected Graduation: Spring 2027
 - Advisor: [Prof. Xuesu Xiao](#)
- **M.S. in Computer Science**, George Mason University (Aug 2019 – May 2021)
 - Advisors: Dr. David Lattanzi and Dr. Amarda Shehu

Research & Work Experience

- **Collaborative Researcher**, Aug 2026 – Present
 - Microsoft Research Asia
 - Collaborators: [Dongsheng Li](#) and [Dongqi Han](#)
- **Research Intern**, May 2026 – Aug 2026
 - Microsoft Research Asia
 - Collaborators: [Dongsheng Li](#) and [Dongqi Han](#)
- **Collaborative Researcher**, Jan 2026 – Present
 - Rutgers University
 - Collaborator: [Chengzhi Mao](#)
- **Collaborative Researcher**, Oct 2025 – Present
 - University of South Florida & University of Maryland
 - Collaborator: [Xiaomin Lin](#) (University of South Florida)
- **Graduate Research Assistant**, Aug 2021 – Present
 - George Mason University, Fairfax, VA
 - Advisor: [Xuesu Xiao](#)
- **Research Engineer Intern**, May 2025 – Aug 2025
 - Johns Hopkins University, Baltimore, MD
 - Collaborator: [Tinoosh Mohsenin](#)
- **Digital Innovation Research Fellow**, May 2024 – Aug 2024
 - Unitree Robotics × Institute for Digital Innovation (IDIA)
 - IDIA fellowship in collaboration with Unitree Robotics.
- **Visiting Researcher (Remote)**, Aug 2023 – May 2024
 - University of Oxford
- **Summer Research Assistant**, May 2020 – Aug 2020
 - Virginia Transportation Research Council (VTRC)
 - In collaboration with George Mason University

Selected Publications

- **Learning Scene-Aware Humanoid Locomotion through 3D Clutter from Immersive Human Demonstrations**
B. Wang, T. Xu, D. Kosukhin, Y.-H. Yeung, **Y. Lu**, X. Xiao — Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027
- **CAVE-NAV: VLM-Based Autonomous 3D Navigation in Underwater Cave Environments**
Z. Wu, **Y. Lu**, Y. Zhang, M. Yu, X. Xiao, J. Shin, X. Lin — Accepted to OCEANS, 2026
- **APPLV: Adaptive Planner Parameter Learning from Vision-Language-Action Model**
Y. Lu, B. Wang, Z. Wu, Y. Li, X. Lin, C. Mao, X. Xiao — Accepted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026
- **CORAL: COntextual Reasoning And Local Planning in A Hierarchical VLM Framework for Underwater Monitoring**
Z. Wu, **Y. Lu**, X. Xiao, X. Lin — Accepted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026
- **Accounting for Work Zone Disruptions in Traffic Flow Forecasting via Multi-Channel Attention-Based Spatio-Temporal Graph Convolutional Networks**
Y. Lu, S. Zhu, A. Shehu, D. Lattanzi — IET Intelligent Transport Systems, 2026
- **Autonomous Ground Navigation in Highly Constrained Spaces: Lessons Learned from the Fourth BARN Challenge at ICRA 2025**
X. Xiao, Z. Xu, S. A. Ghani, A. Datar, D. Song, P. Stone, A. Mazen, K. Yazdipaz, I. Mateyaunga, M. Faied, M. Krishnan, **Y. Lu**, T. Xu, N. Mohammad, W. Kim, J. Reasoner, R. Chandra, N. Bezzo — IEEE Robotics & Automation Magazine, 33(1), 186–196, 2026
- **Adaptive Dynamics Planning for Robot Navigation**
Y. Lu, M. Mao, T. Xu, L. Wang, X. Lin, X. Xiao — Accepted to IEEE International Conference on Robotics and Automation (ICRA), 2026

[Full publication list](#)

Skills

- **Programming:** C++, Python, Java
- **Machine Learning:** PyTorch, TensorFlow, Keras, Scikit-Learn
- **Robotics:** ROS (ROS1/ROS2), Gazebo, Isaac Sim/Lab, OpenAI Gym
- **Other:** Singularity/Apptainer, HPC (Slurm), Docker, Git

Reviewer

Service period: 2024–2026

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Robotics and Automation Letters (RAL)

Leadership

Team Lead, GMU BARN Challenge 2025 Team